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#Assignment 4: Python programming language
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#Dataset
plyaer = [
    ["Serena Williams", "Tennis", 23],
    ["Lionel Messi", "Soccer", 7],
    ["Michael Phelps", "Swimming", 8],
    ["Usan Bolt", "Atheletics", 8],
    ["Roger Federer", "Tennis", 20],
    ["Critiano Ronaldo", "Soccer", 5]
]
```

```
# 1. WRITE A PYTHON PROGRAM TO CHECK IF A PLAYER LIONEL MESSI HAS MORE THAN 10 ACHIEVEMENTS .
# IF THE CONDITION IS TRUE , PRINT PLAYER'S NAME,SPORT,AND ACHIEVEMENTS ELSE PRINT DOES NOT HAVE MORE T
for player_data in plyaer:
    if player_data[0] == "Lionel Messi":
        if player_data[2] > 10:
            print(player_data[0], player_data[1], player_data[2])
        else:
            print("Lionel Messi does not have more than 10 achievements.")
```

```
Lionel Messi does not have more than 10 achievements.
```

```
# 2. WRITE A PYTHON PROGRAM TO CHECK IF A PLAYER BELOGS TO THE SPORT TENNIS OR HAS EXACTLY 20 ACHIEVENT
#IF THE CONDITION IS TRUE , PRINT A SUCCESS MESSAGE.
for player in plyaer:
    if player[1] == "Tennis" or player[2] == 20:
        print(player)
```

```
['Serena Williams', 'Tennis', 23]
['Roger Federer', 'Tennis', 20]
```

```
# 3. WRITE A PYTHON PROGRAM TO CHECK IF A PLAYER HAS LESS THAN 10 ACHIEVEMENTS AND DOES NOT PLAY SOCCER
# PRINT THEIR DETAILS IF THEY MEET THE CRITERIA.
for player in player:
    if player[2] < 10 and player[1] != "Soccer":
        print(player)
```

```
['Michael Phelps', 'Swimming', 8]
['Usan Bolt', 'Atheletics', 8]
```

```
# 4. WRITE A PYTHON PROGRAM IN A JUPYTER NOTEBOOK THAT TAKES AAN INTEGER INPUT (1 TO 9999) FROM THE USE
# DETERMINES THE FOLLOWING USING BRANCING STATEMENTS:
num = int(input("Enter an integer between 1 and 9999: "))

if num < 1 or num > 9999:
    print("enter a number between 1 and 9999.")
else:
    # Check even or odd
    if num % 2 == 0:
        print("The number is even.")
    else:
        print("The number is odd.")

    # Check palindrome
    original = num
    reverse = 0

    while num > 0:
        digit = num % 10
        reverse = reverse * 10 + digit
        num = num // 10

    if original == reverse:
        print("The number is a palindrome.")
    else:
        print("The number is not a palindrome.")

    # Calculate sum of digits
```

```
temp = original
digit_sum = 0

while temp > 0:
    digit_sum = digit_sum + temp % 10
    temp = temp // 10

# Check divisibility
if original % digit_sum == 0:
    print("The number is divisible by the sum of its digits.")
else:
    print("The number is not divisible by the sum of its digits.")
```

Enter an integer between 1 and 9999: 400
The number is even.
The number is not a palindrome.
The number is divisible by the sum of its digits.

```
# 5. WRITE A FOR LOOP THAT PRINTS OUT ALL THE ELEMENTS BETWEEN -5 AND 5 USING THE RANGE FUNCTION.
for i in range(-5, 6):
    print(i)
```

```
-5
-4
-3
-2
-1
0
1
2
3
4
5
```

```
# 6. WRITE A LOOP THAT PRINTS OUT OF THE FOLLOWING LIST: squares = ['red','yellow','green','purple','blue']
squares = ['red', 'yellow', 'green', 'purple', 'blue']

for color in squares:
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print(color)
```

```
red  
yellow  
green  
purple  
blue
```

```
# 7. WRITE A WHILE LOOP TO COPY THE STRINGS 'ORANGE' OF THE LIST TO THE LIST NEW_SQAURES.  
# STOP AND EXIT THE LOOP IF THE VALUE ON THE LIST IS NOT 'ORANGE'  
squares = ['red', 'yellow', 'green', 'purple', 'blue']  
  
new_squares = []  
  
i = 0  
  
while i < len(squares):  
    if squares[i] != 'orange':  
        break  
  
    new_squares.append(squares[i])  
    i += 1  
  
print("New squares:", new_squares)
```

```
New squares: []
```

```
# 8. THE FOLLOWING IS A LIST OF ANIMALS IN A NATIONAL ZOO.  
# ANIMALS = ["LION","GIRAFFE","GORILLA","PARROTS","CROCODILE","DEER","SWAN"]  
# YOUR BROTHER NEEDS TO WRITE AN ESSAY ON THE AMINALS WHOSE NAME ARE MADE OF 7 LETTERS .  
# HELP HIM FIND THOSE ANIMALS THROUGH A WHILE LOOP AND CREATE A SEPARATE LIST OF SUCH ANIMALS.  
Animals = ["lion", "giraffe", "gorilla", "parrots", "crocodile", "deer", "swan"]  
  
seven_letter_animals = []  
  
i = 0
```

```
while i < len(Animals):  
    if len(Animals[i]) == 7:  
        seven_letter_animals.append(Animals[i])  
  
    i += 1  
  
print("Animals with 7 letters:")  
print(seven_letter_animals)
```

```
Animals with 7 letters:  
['giraffe', 'gorilla', 'parrots']
```